Extremal mass at fixed angular momentum in EGB, $D = 5$

Status as of 23 October 2026 (previous version: 9 October). Asymptotically flat black holes, two equal spins $J_1 = J_2 = J$, $G_5 = 1$, action $R + \alpha\mathcal{L}_{\rm GB}$.

We want the smallest mass $M_{\rm ext}(J, \alpha)$ that still has a horizon when J is held fixed and the Gauss–Bonnet coupling α is varied. By dimensional analysis it is a function of one variable:

$$M_{\rm ext} = \pi^{1/3} J^{2/3}\, m(s), \qquad s = \frac{\pi^{2/3}\alpha}{J^{2/3}}.$$

$s \to 0$ is weak coupling (Myers–Perry, $m = 3/2$); $s \to \infty$ is strong coupling, or $J \to 0$ at fixed α.

Summary. We now know $m(s)$ well at both ends and numerically in between:

$$m(s) = \begin{cases} \frac{3}{2} + s - \frac{236}{105}s^2 + d_3 s^3 + \cdots + d_6 s^6 + \ldots, & s \to 0 \text{ (exact coefficients)}, \\ \frac{3}{4}s + \frac{4}{3}s^{-1/2} + a_2 s^{-1} + a_3 s^{-3/2} + \ldots, & s \to \infty, \quad a_2 = 0.4566 \pm 0.0002,\ a_3 \simeq -0.263, \end{cases}$$

and a ten-term formula (§4) that reproduces the numerical solution for all $0 < s \le 103$ to 1×10^{-5}. In physical units the strong-coupling expansion reads

$$M_{\rm ext} = \frac{3\pi\alpha}{4} + \frac{4J}{3\sqrt{\alpha}} + a_2 \frac{J^{4/3}}{\pi^{1/3}\alpha} + a_3 \frac{J^{5/3}}{\pi^{2/3}\alpha^{3/2}} + \ldots$$

No closed form for $m(s)$ was found, and there are reasons (below) to think none of a simple kind exists.

1. Weak coupling: the series through s^6. Same method as before: the exterior metric of the extremal hole is solved exactly order by order in α (multiple polylogarithms in $1/r^2$), and the mass is read off at infinity. Two new orders:

$$d_3 = -6.548307, \quad d_4 = 66.100522, \quad d_5 = 423.920197004910\ldots, \quad d_6 = -4737.494826989936\ldots$$

(exact forms, with $\zeta(3), \zeta(5), \zeta(3)^2, \pi^{2k}$, in `series/coefficients.json` and in `strong-coupling-report.pdf`). Checks: the extremal first law $dM_{\rm ext} = 2\Omega_H\, dJ$ holds with zero residual at orders 2–6, and the horizon radius at orders 5 and 6 matches an independent prediction (next paragraph) exactly. Same caveat as before: exactness rests on identifying certain horizon constants with zeta values to 100 digits.

The series has a small radius of convergence. The near-horizon geometry of the extremal hole turns out to be fixed by algebraic equations, so the horizon size is an explicit algebraic function of s. That function has its nearest singularities at $s_* = 0.0059 \pm 0.0548\, i$, i.e. $|s_*| = 0.055$. The signs of $d_1, \ldots, d_6$ follow the pattern this singularity predicts (6 of 6). Our conclusion, not proved: the weak series converges only for $|s| \lesssim 0.055$ and is useful only as an asymptotic expansion; a resummation that respects s_* follows the numerical $m(s)$ up to $s \approx 0.5$, converging slowly.

2. Strong coupling: the leading terms. *(a) Where 4/3 comes from.* We followed the suggestion to use the pure-Lovelock solutions of Dadhich, Pons and Prabhu [3]. In $D = 5$, EGB is the Lovelock theory they call "derivative-degenerate", and its small static black holes ($r \ll \sqrt{\alpha}$) behave like pure Gauss–Bonnet in its critical dimension, which has the thermodynamics of a three-dimensional BTZ black hole. We derived the dictionary: AdS radius $\ell^2 = 4\alpha$ and $1/(4G_3) = 3\pi\alpha$. The static hole with zero horizon radius, of mass $3\pi\alpha/4$, is the massless BTZ point. For an extremal BTZ hole the mass above that point is $\Delta M = S^2/(48\pi^3\alpha^2)$ (standard BTZ relation, recalled). Inserting the exact entropy of the extremal hole from the algebraic near-horizon equations gives $\Delta M \to 4J/(3\sqrt{\alpha})$, i.e. $a_1 = 4/3$. This is the same coefficient an earlier, independent matched-asymptotics argument had produced. The step that is assumed rather than derived: the rotating extremal core obeys the static BTZ dictionary.

(b) The numbers agree. We solved the full field equations up to $s = 103$ (the earlier numerical branch stopped at $s = 5.24$) and fitted $(m - 3s/4)\sqrt{s} = a_1 + a_2 s^{-1/2} + a_3 s^{-1} + \ldots$:

- all coefficients free: $a_1 = 1.33333$ (fits of different length: 1.3333–1.3337);
- with $a_1 = 4/3$ fixed: $a_2 = 0.45658$–0.45669. The earlier matched analysis had predicted $a_2 = 0.456527$ under extra assumptions; fixing that value fits the data as well as leaving it free;
- with a_1, a_2 fixed: a_3 between -0.2625 and -0.2623 (with a_2 free: -0.268 to -0.265).

So $a_1 = 4/3$ is confirmed to about 3×10^{-4}, a_2 is consistent with the predicted value to about 10^{-4}, and a_3 is a new, purely numerical result.

3. The numerical solution. Up to $s \approx 9$ ordinary methods work (shooting from both ends, checked against the earlier spectral solution). Beyond that, a deformation of the horizon S^3 (squashing) behaves like $e^{\pm 4\sqrt{6}\,\alpha/r}$ inside the core (a factor e^{76} already at $\alpha = 10$), so integrating from one end is hopeless. We used multiple shooting: about 50–130 short segments, matched by Newton's method, with quadruple precision near the horizon (where the

solution has a non-analytic piece $\propto x^\gamma$, γ known exactly) and the large-r series at the outer end. One lesson worth recording: our first run put the outer end at $25\sqrt{\alpha}$; there the far-field information is buried below double-precision noise and the masses came out wrong by up to 10^{-6}. That run was discarded and redone at $12\sqrt{\alpha}$ and $8\sqrt{\alpha}$, which agree with each other to 5×10^{-10}. Selected values:

s	2	2.982	9.74	20.28	48.87	102.86
$m(s)$	2.4446986	3.0410067	7.7627041	15.5223944	36.8490546	77.2785159

4. A formula for all s. With $t = (1 + s/\kappa)^{-1/2}$ and $\kappa = 0.356242695$,

$$m(s) \approx \frac{3s}{4} + \sum_{k=1}^{10} c_k\, t^k,$$

$$c_{1\ldots5} = 2.233910,\ 1.281505,\ -0.713812,\ 6.055449,\ -75.189708,$$
$$c_{6\ldots10} = 230.148117,\ -352.252482,\ 303.906361,\ -142.214324,\ 28.244984.$$

It has the right structure at both ends (a Taylor series at $s = 0$, a series in $s^{-1/2}$ at infinity), reproduces $m(0) = 3/2$, $m'(0) = 1$, $a_1 = 4/3$, $a_2 = 0.456527$ by construction, and matches all 126 numerical points to 1.1×10^{-5} (relative; checked with the printed coefficients). It is a fit, not the exact function: it misses the singularity at $|s_*| = 0.055$, so it should not be expanded beyond first order in s.

5. The point $J = 0$. The leading term $3\pi\alpha/4$ is the mass of the static solution with zero horizon radius, which is singular; the extremal rotating branch ends there, as Kleihaus, Kunz and Radu conjectured [2]. Our numerics now follow the branch to $J/\alpha^{3/2} = \pi s^{-3/2} \approx 0.003$ and show it approaching this point as the expansion above says.

6. What is established and what is not. *Exact:* $d_1, \ldots, d_6$ (given the zeta identifications); the near-horizon data as algebraic functions of s; the location of s_*. *Derived under one assumption:* $a_1 = 4/3$ (BTZ dictionary for the rotating core). *Numerical:* $m(s)$ for $0 < s \le 103$, a_1 to 3×10^{-4}, a_2, a_3. *Not proved:* that s_* is the nearest singularity of $m(s)$ itself; the analytic value of a_2. *Not found:* a closed form. The zeta values in d_n rule out an algebraic $m(s)$ with rational coefficients, and the complex singularity at $|s| = 0.055$ makes an elementary formula built from the two ends unlikely.

Possible next steps. Derive a_2 properly (the next order of the core/BTZ matching); push the numerics to $s \sim 200$ (about 6–8 hours with the present code) to pin down a_3.

[1] Ma, Li, Lü, arXiv:2009.00015. [2] Kleihaus, Kunz, Radu, arXiv:2303.12471. [3] Dadhich, Pons, Prabhu, arXiv:1201.4994.
Technical details and all checks: `strong-coupling-report.pdf`. Code and outputs: `extremal-mass/code/`, `series/`, `strong/`; restart notes `RESUME.txt`.